Slides for Patryk's Notes

Patryk Kozlowski

October 7, 2026

Outline

1. Madelung treatment

2. Derivation of the quasiboson Hamiltonian

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2. Derivation of the quasiboson Hamiltonian

Derivation from HF total energy

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$$\mathrm{IP} \equiv E_{\mathrm{HF}}(N-1) - E_{\mathrm{HF}}(N), \quad \mathrm{EA} \equiv \mathrm{E_{HF}(N+1) - E_{HF}(N)} \tag{3}$$

$$= -\left(\frac{i^2}{2} + \Sigma_i^x \right) + \frac{|v_M|}{2}, \quad = \left(\frac{a^2}{2} + \Sigma_a^x \right) - \frac{|v_M|}{2} \tag{4}$$

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since $E_x(N-1) = E_x(N) - \Sigma_i^x$ and $E_x(N+1) = E_x(N) + \Sigma_a^x$.

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I interpret what you write after equation 64 as saying that it makes no difference whether the Madelung constant gets lumped into the single-particle energies, as in Eq. (2), or added afterwards as in Eq. (1). I interpret what you write after equation 66 as saying that

$$E_g \equiv EA + \frac{|v_M|}{2} + IP + \frac{|v_M|}{2} \tag{5}$$

$$= \left(\frac{j^2}{2} + \Sigma_j^x - \frac{|v_M|}{2}\right) + \left[-\left(\frac{i^2}{2} + \Sigma_i^x\right) + \frac{|v_M|}{2}\right] + |v_M| \tag{6}$$

$$= \frac{j^2}{2} + \Sigma_j^x - \frac{i^2}{2} - \Sigma_i^x + |v_M| \tag{7}$$

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$$E_{\mathrm{HF}}(N) = \sum_{\mathbf{k}} \frac{k^2}{2} n_k - \frac{1}{2} \sum_{\mathbf{k}} n_{\mathbf{k}} \left(\frac{1}{\Omega} \sum_{\mathbf{q}} v_q \, n_{\mathbf{k}+\mathbf{q}} \right) \tag{8}$$

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$$= \sum_{\mathbf{k}} \frac{k^2}{2} n_k + \frac{1}{2} \sum_{\mathbf{k}} \Sigma_k^x \, n_{\mathbf{k}} - \frac{|v_M|}{2} \sum_{\mathbf{k}} n_{\mathbf{k}}^2 \tag{10}$$

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Recall that HF energies came out to be $\varepsilon_k^{\mathrm{HF}} = \frac{k^2}{2} + \Sigma_k^x - |v_M| \, n_{\mathbf{k}}$.

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$$E^{(2)} = -\sum_{ijab} \frac{\mathcal{K}^{ab}_{ij}}{\tilde{E}^{(0)}_{ij\to ab}} \tag{12}$$

$$\tilde{E}^{(0)}_{ij\to ab} = E^{(0)}_{ij\to ab} + 2|v_M| \tag{13}$$

$$\Sigma^{(2)}_{pp}(\omega) = \sum_{ija} \frac{\mathcal{K}^{ap}_{ij}}{\omega - \epsilon^i + (\epsilon_a - \epsilon_j) - 2|v_M|} + \sum_{iab} \frac{\mathcal{K}^{ab}_{ip}}{\omega - \epsilon_a - (\epsilon_b - \epsilon_i) + |v_M|} \tag{14}$$

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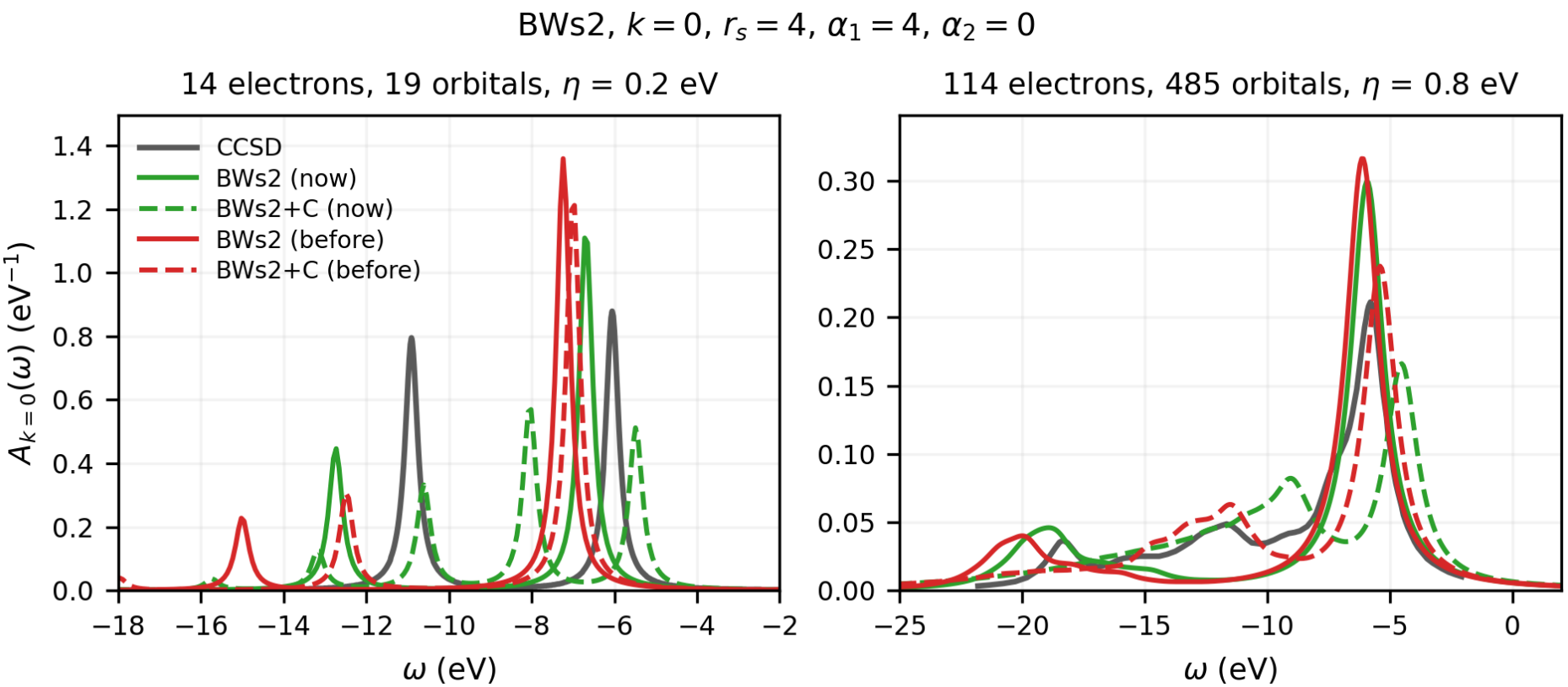
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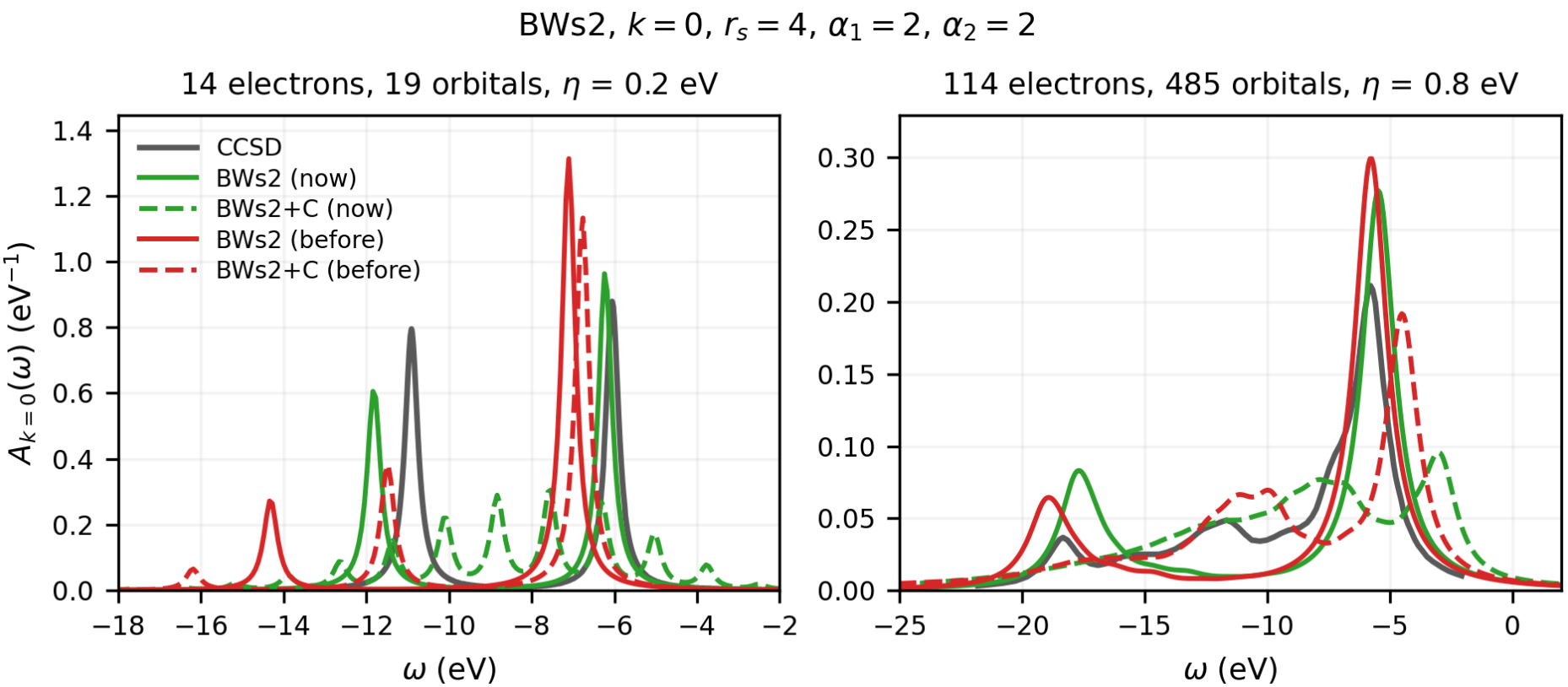
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Updated results: BW 4,0



Updated results: BW 2,2



How HF+GW changes

1. Calculate the RPA polarizability

$$\begin{pmatrix} A & B \\ B & A \end{pmatrix} \begin{pmatrix} X & Y \\ Y & X \end{pmatrix} = \begin{pmatrix} X & -Y \\ -Y & X \end{pmatrix} \begin{pmatrix} \Omega & 0 \\ 0 & \Omega \end{pmatrix} \tag{17}$$

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2. Form the correlation self-energy as

$$\Sigma_{pp}^{c,G_0W_0}(\omega) = \sum_{j\nu} \frac{W_{pj\nu} W_{pj\nu}}{\omega - (\epsilon_j - \Omega_\nu) - \mathrm{i}\eta} + \sum_{b\nu} \frac{W_{pb\nu} W_{pb\nu}}{\omega - (\epsilon_b + \Omega_\nu) + \mathrm{i}\eta}, \tag{18}$$

where $W_{np\nu} = \sqrt{2} \sum_{ia} V_{npia} (\mathbf{X} + \mathbf{Y})_{ia,\nu}$ with $V_{npia} = (ni|pa)$.

How HF+GW changes cont.

Before :

$$
\begin{aligned}
A_{ia,jb} &= \delta_{ij}\delta_{ab}\left(\epsilon_a - \epsilon_i + |v_M|\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle \\
&= \delta_{ij}\delta_{ab}\left(\epsilon_a - \epsilon_i\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle \\
B_{ia,jb} &= \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle
\end{aligned}
$$

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How HF+GW changes cont.

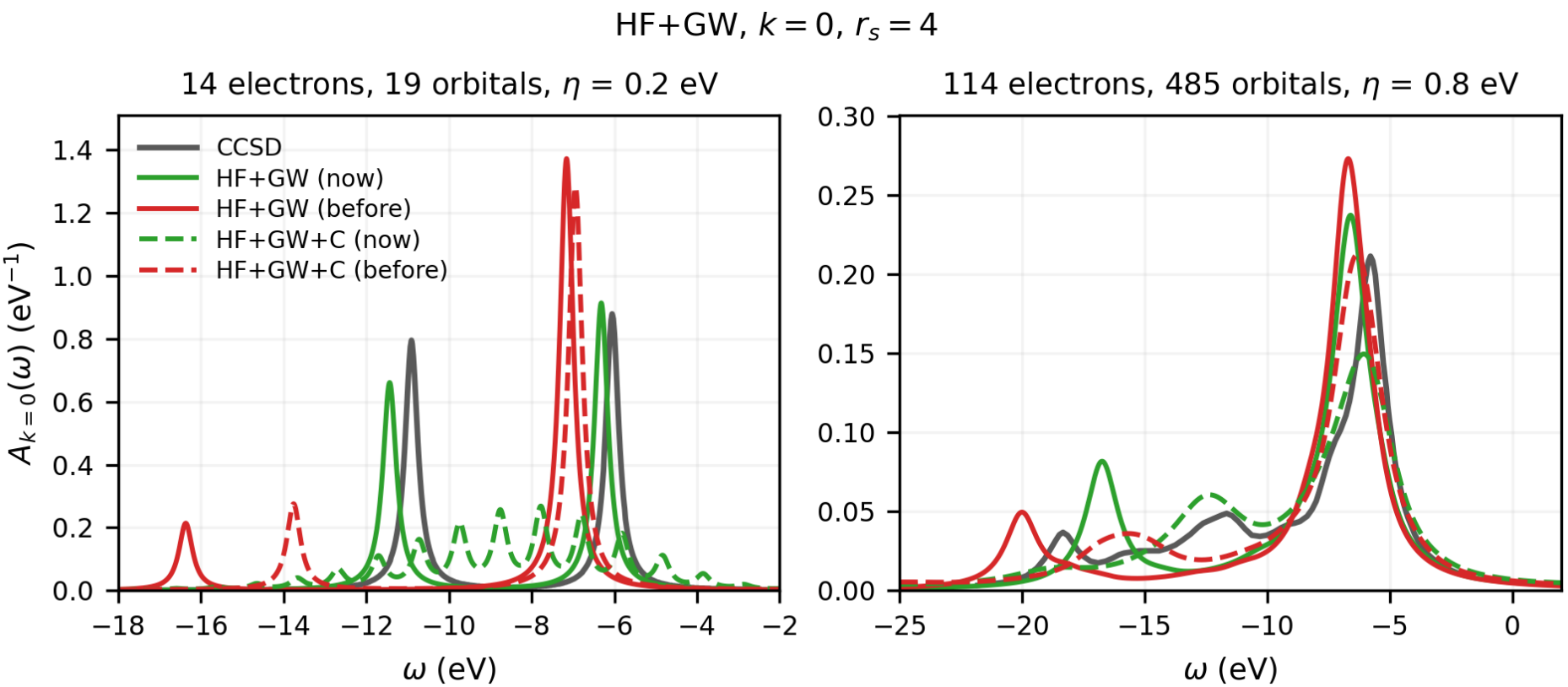
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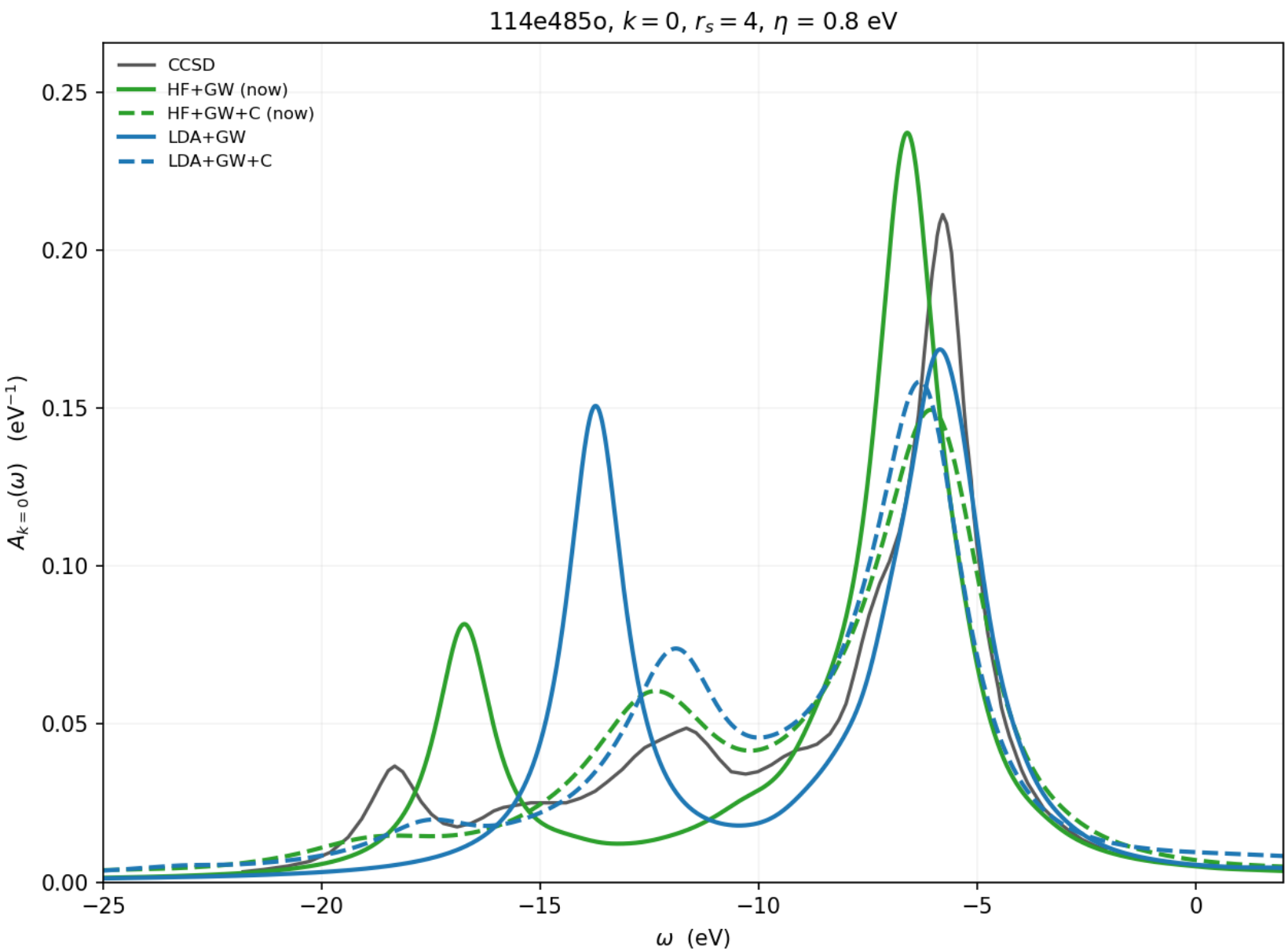
Now :

$$A_{ia,jb} = \delta_{ij}\delta_{ab}\left(\epsilon_a - \frac{|v_M|}{2} - \epsilon_i + \frac{|v_M|}{2}\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle \quad (19)$$
$$= \delta_{ij}\delta_{ab}\left(\epsilon_a - \epsilon_i\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle \quad (20)$$
$$B_{ia,jb} = \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle \quad (21)$$
$$\Sigma_{pp}^{c,G_0W_0}(\omega) = \sum_{j\nu} \frac{W_{pj\nu} W_{pj\nu}}{\omega - \left(\epsilon_j + \frac{v_M}{2} - \Omega_\nu\right) + \mathrm{i}\eta} + \sum_{b\nu} \frac{W_{pb\nu} W_{pb\nu}}{\omega - \left(\epsilon_b + \frac{v_M}{2} + \Omega_\nu\right) + \mathrm{i}\eta} \quad (22)$$

Updated results: HF+GW



Comparing updated HF+GW to LDA+GW



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3. Redo all TDL extrapolations with Madelung correction.

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1. Madelung treatment

2. Derivation of the quasiboson Hamiltonian

The goal

With $\epsilon_{\mathbf{k}}^{(0)} = \frac{k^2}{2} + \frac{v_M}{2}$ and $v_M = -2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_s^{-1}$

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{(0)} \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{k}} + \frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{k}\mathbf{k}'} a_{\mathbf{k}+\mathbf{q}}^{\dagger} a_{\mathbf{k}'-\mathbf{q}}^{\dagger} a_{\mathbf{k}'} a_{\mathbf{k}} \tag{23}$$

$$= E_{\mathrm{HF}} + \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} : \hat{a}_{\mathbf{k}}^{\dagger} \hat{a}_{\mathbf{k}} : + \frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{k}\mathbf{k}'} : a_{\mathbf{k}+\mathbf{q}}^{\dagger} a_{\mathbf{k}'-\mathbf{q}}^{\dagger} a_{\mathbf{k}'} a_{\mathbf{k}} : \tag{24}$$

where $\epsilon_{\mathbf{k}}^{\mathrm{HF}} = \frac{k^2}{2} + \frac{v_M}{2} - \frac{1}{\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q n_{\mathbf{k}+\mathbf{q}}$.

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$$\hat{H}_{\mathrm{eb}} = -\sum_{\mathbf{k}\in\mathcal{O}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} h_{\mathbf{k}}^\dagger h_{\mathbf{k}} + \sum_{\nu_{\mathbf{q}}} \omega_{\nu_{\mathbf{q}}}^{\mathrm{pl}} \hat{\beta}_{\nu_{\mathbf{q}}}^\dagger \hat{\beta}_{\nu_{\mathbf{q}}} + \sum_{\mathbf{k}\mathbf{q}} \sum_{\nu_{\mathbf{q}}} g_\nu^{\mathrm{pl}}(\mathbf{k},\mathbf{q}) h_{\mathbf{k}+\mathbf{q}}^\dagger h_{\mathbf{k}} \left(\hat{\beta}_{\nu_{\mathbf{q}}} + \hat{\beta}_{\nu_{-\mathbf{q}}}^\dagger\right). \tag{25}$$

Quadratic bosonic bath

Define $E^{\dagger}_{i\mathbf{q}} = a^{\dagger}_{\mathbf{i}+\mathbf{q}} a_{\mathbf{i}}$ and $E_{i-q} = a^{\dagger}_{\mathbf{i}} a_{\mathbf{i}-\mathbf{q}}$ with $\hat{\kappa}[\mathbf{z}] = \sum_{i\mathbf{q}} \left[\kappa_{i\mathbf{q}} E^{\dagger}_{i\mathbf{q}} - \kappa^{*}_{i\mathbf{q}} E_{i\mathbf{q}} \right]$

using $\boldsymbol{\kappa}_{\mathbf{q}} = \begin{pmatrix} \kappa_{1\mathbf{q}} \\ \kappa_{2\mathbf{q}} \\ \vdots \end{pmatrix}$ and $\mathbf{z}_{\mathbf{q}} = \begin{pmatrix} \boldsymbol{\kappa}_{\mathbf{q}} \\ \boldsymbol{\kappa}^{*}_{-\mathbf{q}} \end{pmatrix}$.

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$$E(\mathbf{z}) = \langle \Phi(\mathbf{z}) | \hat{H} | \Phi(\mathbf{z}) \rangle \tag{26}$$

$$= E_{\mathrm{HF}} + \langle \Phi_0 | [\hat{H}, \hat{\kappa}[\mathbf{z}]] | \Phi_0 \rangle + \frac{1}{2} \langle \Phi_0 | [[\hat{H}, \hat{\kappa}[\mathbf{z}]], \hat{\kappa}[\mathbf{z}]] | \Phi_0 \rangle + \dots \tag{27}$$

$$\implies \nabla_{\mathbf{z}} E(\mathbf{z}) = \begin{pmatrix} \dfrac{\partial E(\mathbf{z})}{\partial \boldsymbol{\kappa}} \\ \\ \dfrac{\partial E(\mathbf{z})}{\partial \boldsymbol{\kappa}^{*}} \end{pmatrix} \implies \nabla_{\mathbf{z}} E(\mathbf{z})\big|_{\mathbf{z}=0} = 0 \tag{28}$$

Quadratic bosonic bath cont.

$$\mathcal{H}_{\mathrm{HF}} = \left.\frac{\partial^2 E(\mathbf{z})}{\partial \begin{pmatrix} \boldsymbol{\kappa}^* \\ \boldsymbol{\kappa} \end{pmatrix} \partial \begin{pmatrix} \boldsymbol{\kappa} & \boldsymbol{\kappa}^* \end{pmatrix}}\right|_{\mathbf{z}=0} = \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}. \tag{29}$$

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Thus, the second-order correction to the HF energy is

$$E^{(2)}(\kappa, \kappa^*) = \frac{1}{2} \begin{pmatrix} \boldsymbol{\kappa}^*_{\mathbf{q}} & \boldsymbol{\kappa}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} \boldsymbol{\kappa}_{\mathbf{q}} \\ \boldsymbol{\kappa}^*_{-\mathbf{q}} \end{pmatrix} \tag{30}$$

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Quantizing the space and then applying Bogoliubov transformation gives

$$\hat{H}^{(2)} = \frac{1}{2} \begin{pmatrix} \hat{\boldsymbol{b}}^\dagger_{\mathbf{q}} & \hat{\boldsymbol{b}}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{b}}_{\mathbf{q}} \\ \hat{\boldsymbol{b}}^\dagger_{-\mathbf{q}} \end{pmatrix} \tag{31}$$

$$= \hat{\boldsymbol{\beta}}^\dagger_{\mathbf{q}} \boldsymbol{\Omega}_{\mathbf{q}} \hat{\boldsymbol{\beta}}_{\mathbf{q}} + \frac{1}{2} \mathrm{Tr}(\boldsymbol{\Omega}_{\mathbf{q}} - A) + \frac{1}{2} \mathrm{Tr}(A) \tag{32}$$

$$= \sum_{\nu} \omega_{\nu_{\mathbf{q}}} \hat{\beta}^\dagger_{\nu_{\mathbf{q}}} \hat{\beta}_{\nu_{\mathbf{q}}} + E_{\mathrm{RPA}} + \frac{1}{2} \mathrm{Tr}(A) \tag{33}$$

Pure fermionic and coupling terms

Start with

$$\hat{H}_{\mathrm{h}}(\boldsymbol{\kappa}) = \sum_{\mathbf{k},\mathbf{k}' \in \mathcal{O}} h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}(\boldsymbol{\kappa}) h^{\dagger}_{\mathbf{k}'} h_{\mathbf{k}}. \tag{34}$$

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Expand to first order about the Hartree–Fock reference

$$h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}(\boldsymbol{\kappa}) = h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}(0) + \sum_{i\mathbf{q}} \left.\frac{\partial h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 \kappa_{i\mathbf{q}} + \sum_{i\mathbf{q}} \left.\frac{\partial h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}}{\partial \kappa^{*}_{i\mathbf{q}}}\right|_0 \kappa^{*}_{i\mathbf{q}} + \mathcal{O}(\kappa^2), \tag{35}$$

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$$\left.\frac{\partial h^{\mathrm{h}}_{\mathbf{k}+\mathbf{q},\mathbf{k}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 = -\left.\frac{\partial F_{\mathbf{k},\mathbf{k}+\mathbf{q}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 \tag{36}$$

$$= -\sum_{\mathbf{l}} \frac{\partial F_{\mathbf{k},\mathbf{k}+\mathbf{q}}}{\partial P_{\mathbf{l}+\mathbf{q},\mathbf{l}}} \left.\frac{\partial P_{\mathbf{l}+\mathbf{q},\mathbf{l}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 \tag{37}$$

$$= -\frac{v(\mathbf{q})}{\Omega} \sum_{\mathbf{l}} \delta_{\mathbf{l},\mathbf{i}} \tag{38}$$

Pure fermionic and coupling terms cont.

So up to linear order in $\boldsymbol{\kappa}$

$$\hat{H}_{\mathrm{h}}(\boldsymbol{\kappa}) \overset{\text{direct}}{=} -\sum_{\mathbf{k}\in\mathcal{O}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} h_{\mathbf{k}}^{\dagger} h_{\mathbf{k}} - \sum_{\mathbf{kq}} \sum_{i} \frac{v(\mathbf{q})}{\Omega} h_{\mathbf{k}+\mathbf{q}}^{\dagger} h_{\mathbf{k}} \left(\kappa_{i\mathbf{q}} + \kappa_{i,-\mathbf{q}}^{*}\right) \tag{39}$$

Pure fermionic and coupling terms cont.

So up to linear order in $\boldsymbol{\kappa}$

$$\hat{H}_{\mathrm{h}}(\boldsymbol{\kappa}) \overset{\text{direct}}{=} -\sum_{\mathbf{k}\in\mathcal{O}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} h_{\mathbf{k}}^{\dagger} h_{\mathbf{k}} - \sum_{\mathbf{kq}} \sum_{i} \frac{v(\mathbf{q})}{\Omega} h_{\mathbf{k}+\mathbf{q}}^{\dagger} h_{\mathbf{k}} \left(\kappa_{i\mathbf{q}} + \kappa_{i,-\mathbf{q}}^{*}\right) \tag{39}$$

Quantizing again and then applying the Bogoliubov transformation gives

$$\hat{H}_{\mathrm{h-pl}} = \sum_{\mathbf{kq}} \sum_{\nu_{\mathbf{q}}} g_{\nu}^{\mathrm{pl}}(\mathbf{k},\mathbf{q}) h_{\mathbf{k}+\mathbf{q}}^{\dagger} h_{\mathbf{k}} \left(\hat{\beta}_{\nu_{\mathbf{q}}} + \hat{\beta}_{\nu_{-\mathbf{q}}}^{\dagger}\right), \tag{40}$$

Pure fermionic and coupling terms cont.

So up to linear order in $\boldsymbol{\kappa}$

$$\hat{H}_{\mathrm{h}}(\boldsymbol{\kappa}) \overset{\text{direct}}{=} -\sum_{\mathbf{k}\in\mathcal{O}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} h_{\mathbf{k}}^{\dagger} h_{\mathbf{k}} - \sum_{\mathbf{kq}} \sum_{i} \frac{v(\mathbf{q})}{\Omega} h_{\mathbf{k}+\mathbf{q}}^{\dagger} h_{\mathbf{k}} \left(\kappa_{i\mathbf{q}} + \kappa_{i,-\mathbf{q}}^{*}\right) \tag{39}$$

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with

$$g_{\nu}^{\mathrm{pl}}(\mathbf{k},\mathbf{q}) = -\frac{v(\mathbf{q})}{\Omega} \sum_{\mathbf{i}} \left[X_{i\nu}(\mathbf{q}) + Y_{i\nu}(\mathbf{q})\right]. \tag{41}$$

Pure fermionic and coupling terms cont.

So up to linear order in $\boldsymbol{\kappa}$

$$\hat{H}_{\mathrm{h}}(\boldsymbol{\kappa}) \overset{\text{direct}}{=} -\sum_{\mathbf{k}\in\mathcal{O}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} h_{\mathbf{k}}^{\dagger} h_{\mathbf{k}} - \sum_{\mathbf{kq}} \sum_{i} \frac{v(\mathbf{q})}{\Omega} h_{\mathbf{k}+\mathbf{q}}^{\dagger} h_{\mathbf{k}} \left(\kappa_{i\mathbf{q}} + \kappa_{i,-\mathbf{q}}^{*}\right) \tag{39}$$

Quantizing again and then applying the Bogoliubov transformation gives

$$\hat{H}_{\mathrm{h-pl}} = \sum_{\mathbf{kq}} \sum_{\nu_{\mathbf{q}}} g_{\nu}^{\mathrm{pl}}(\mathbf{k},\mathbf{q}) h_{\mathbf{k}+\mathbf{q}}^{\dagger} h_{\mathbf{k}} \left(\hat{\beta}_{\nu_{\mathbf{q}}} + \hat{\beta}_{\nu_{-\mathbf{q}}}^{\dagger}\right), \tag{40}$$

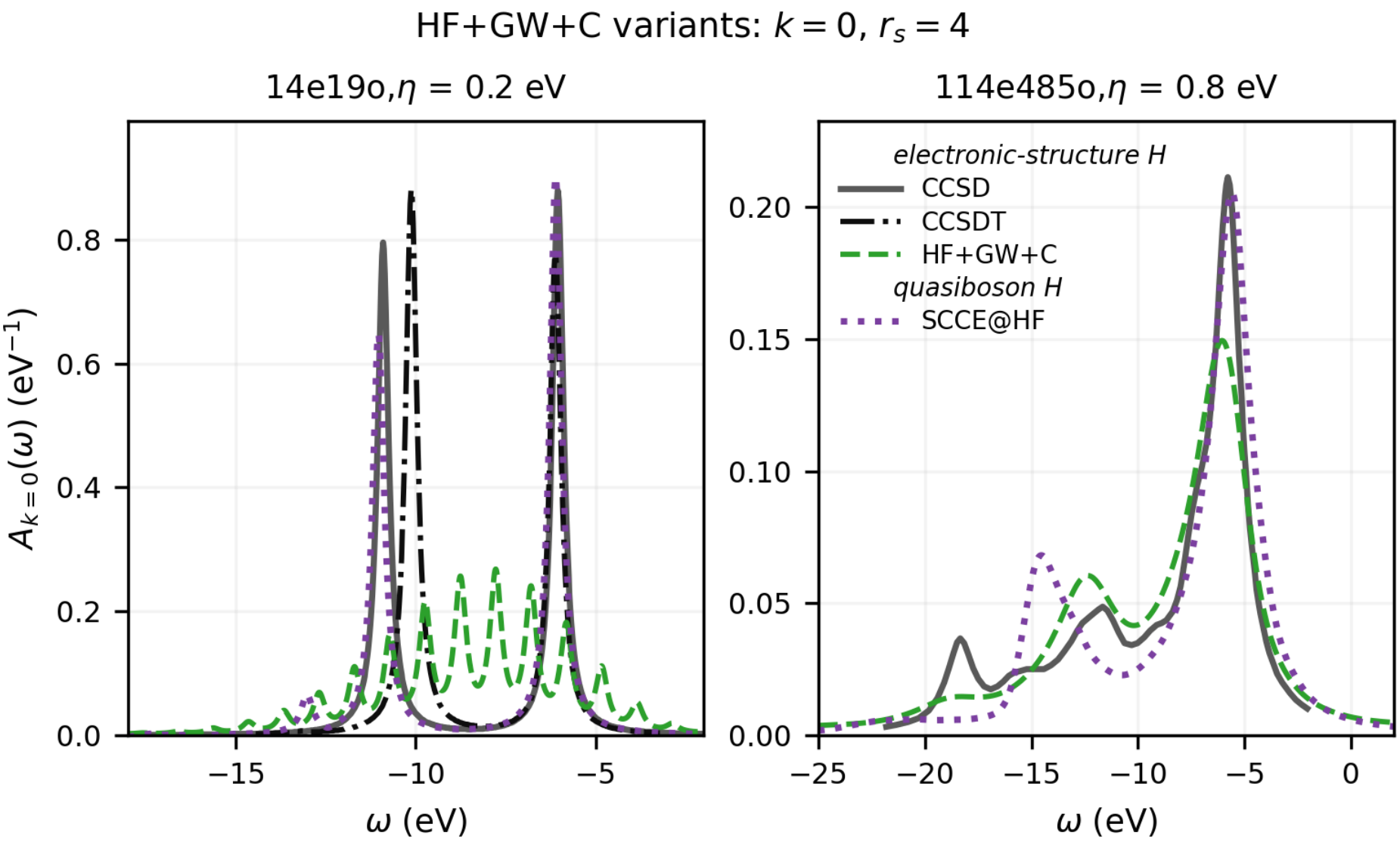
with

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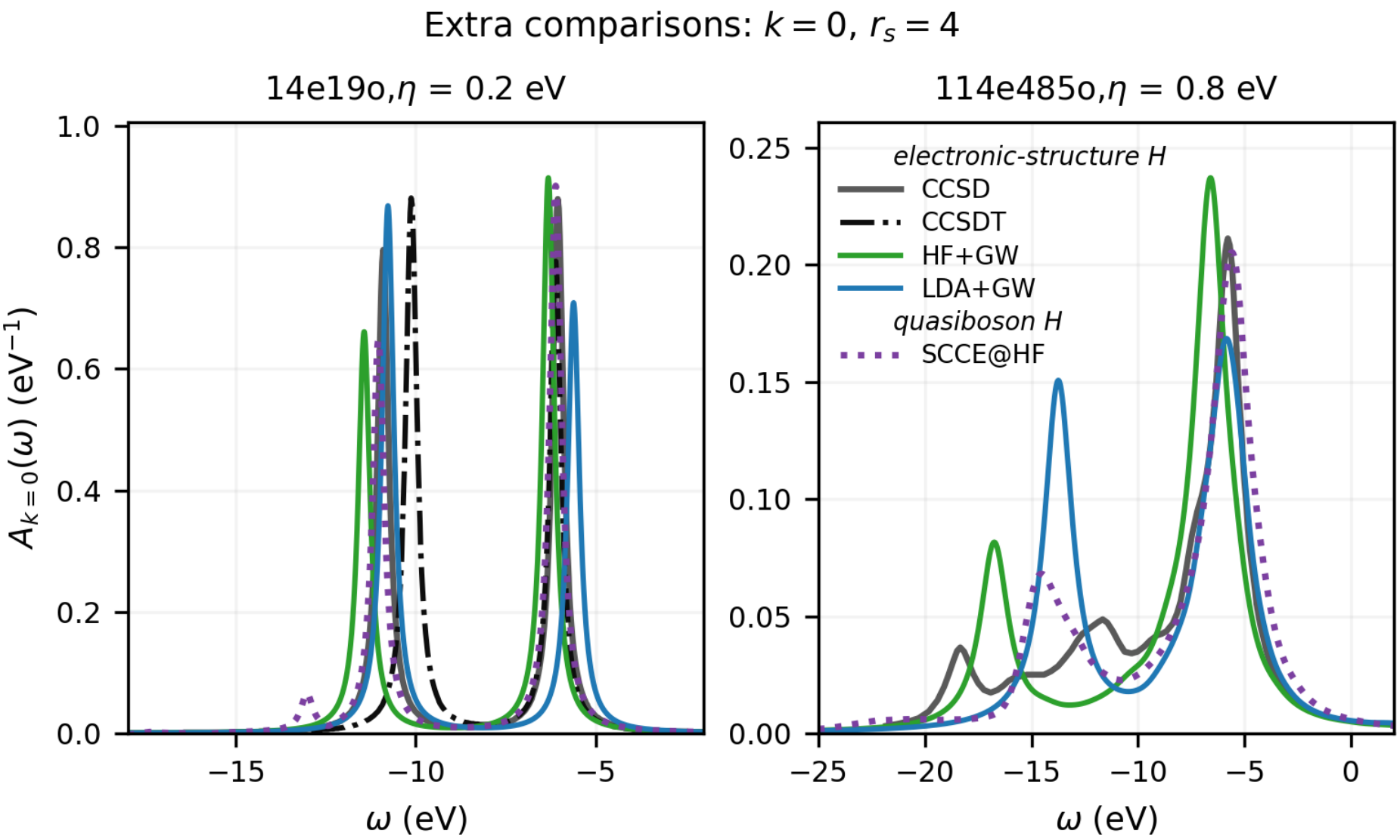
So overall (omitting constant terms)

$$\hat{H}_{\mathrm{eb}} = -\sum_{\mathbf{k}\in\mathcal{O}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} h_{\mathbf{k}}^{\dagger} h_{\mathbf{k}} + \sum_{\nu_{\mathbf{q}}} \omega_{\nu_{\mathbf{q}}}^{\mathrm{pl}} \hat{\beta}_{\nu_{\mathbf{q}}}^{\dagger} \hat{\beta}_{\nu_{\mathbf{q}}} + \sum_{\mathbf{kq}} \sum_{\nu_{\mathbf{q}}} g_{\nu}^{\mathrm{pl}}(\mathbf{k},\mathbf{q}) h_{\mathbf{k}+\mathbf{q}}^{\dagger} h_{\mathbf{k}} \left(\hat{\beta}_{\nu_{\mathbf{q}}} + \hat{\beta}_{\nu_{-\mathbf{q}}}^{\dagger}\right). \tag{42}$$

Results


HF+GW+C variants: $k = 0$, $r_s = 4$
14e19o,η = 0.2 eV
114e485o,η = 0.8 eV
$A_{k=0}(\omega)$ (eV^{-1})
ω (eV)
ω (eV)
electronic-structure H
CCSD
CCSDT
HF+GW+C
quasiboson H
SCCE@HF
0.0
0.2
0.4
0.6
0.8
0.00
0.05
0.10
0.15
0.20
−15
−10
−5
−25
−20
−15
−10
−5
0

Results


Extra comparisons: $k = 0$, $r_s = 4$
14e19o,η = 0.2 eV
114e485o,η = 0.8 eV
$A_{k=0}(\omega)$ (eV^{-1})
ω (eV)
ω (eV)
electronic-structure H
CCSD
CCSDT
HF+GW
LDA+GW
quasiboson H
SCCE@HF